CS432: Databases
Formal Relational Query Languages

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Formal Relational Query Languages: Objective

- To understand formal query languages through relational algebra

Formal Relational Query Languages

- **Background of Relational Model**
- **Relational Algebra**
- Tuple Relational Calculus
- Domain Relational Calculus
- Equivalence of Algebra and Calculus

Background of Relational Model

- Database schema – is the logical structure of the database.
- Database instance – is a snapshot of the data in the database at a given instant in time.
- Example:
  - Schema: `instructor (ID, name, dept_name, salary)`
  - Instance: (shown in table with attributes as columns and tuples as rows)
- A set of allowed values for each attribute called the **domain** of the attribute e.g., alphanumeric string, number, date, null, etc.

attributes
(or columns)

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

tuples
(or rows)

Background of Relational Model (contd..)

- Relation Schema and Instance
  - $A_1, A_2, \ldots, A_n$ are attributes
  - $R = (A_1, A_2, \ldots, A_n)$ is a relation schema
  - Example of an Instructor Relation: `instructor = (ID, name, dept_name, salary)`
  - A relation instance r defined over schema R is denoted by $r(R)$
  - The current values of a relation are specified by a table
  - An element t of relation r is called a tuple and is represented by a row in a table
  - Order of rows/tuples is irrelevant
  - No two tuples are identical

attributes
(or columns)

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
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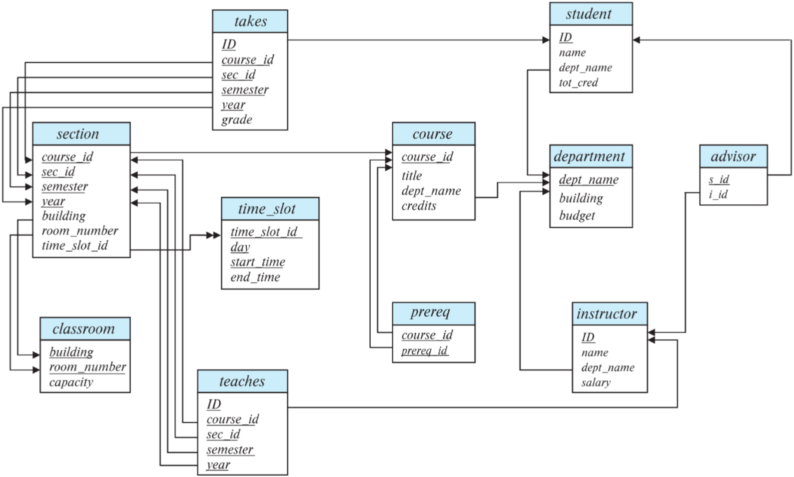
98345	Kim	Elec. Eng.	80000

tuples
(or rows)

Background of Relational Model (contd..)

- Schema Diagram for University Database



Formal Relational Query Languages

- Relational Algebra
  - Procedural and Algebra based
- Tuple Relational Calculus
  - Non-Procedural and Predicate Calculus based
- Domain Relational Calculus
  - Non-Procedural and Predicate Calculus based

Relational Algebra

- Introduced by Edgar F. Codd at IBM in 1970
- Procedural language
- Six basic operators:
  - Select: σ
  - Project: Π
  - Union: $\cup$
  - Set difference: $-$
  - Cartesian product: $\times$
  - Rename: ρ
- The operators take one or two relations as inputs and produce a new relation as a result

Select Operation

- Notation: $\sigma_p(r)$
- p is called the **selection predicate**
- Defined as:

$$\sigma_p(\mathbf{r}) = \{t \mid t \in r \textbf{ and } p(t)\}$$

Where p is a formula in propositional calculus consisting of terms connected by: $\wedge$ (**and**), $\vee$ (**or**), $\neg$ (**not**)

Each **term** is one of:

```
<attribute> op <attribute> or <constant>
```

where op is one of: $=, \neq, >, \geq, <, \leq$

Select Operation

Example of the selection: Query

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

$$\sigma_{\text{dept\_name}=\text{"}Physics\text{"}}(\text{instructor})$$

Result?

Select Operation

Example of the selection: Query

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
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$\sigma_{\text{dept\_name}=\text{"}Physics\text{"}}(\text{instructor})$

Result?

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
33456	Gold	Physics	87000

Select Operation

Example of the selection: Query

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

$$\sigma_{A=B \wedge D>5}(r)$$

Result?

Select Operation

Example of the selection: Query

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

$$\sigma_{A=B \wedge D>5}(r)$$

Result?

A	B	C	D
α	α	1	7
β	β	23	10

Project Operation

- Notation:

$$\Pi_{A_1, A_2, \ldots, A_k}(r)$$

  where A_1, A_2 are attribute names and r is a relation name
- The result is defined as the relation of k columns obtained by erasing the columns that are not listed
- Duplicate rows removed from result, since relations are sets

Project Operation

Example of the Projection: Query

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
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15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

$$\Pi_{ID,\ name,\ salary}(r)$$

Result?

Project Operation

Example of the Projection: Query

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
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$$\Pi_{ID,\ name,\ salary}(r)$$

Result?

ID	*name*	*salary*
10101	Srinivasan	65000
12121	Wu	90000
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76543	Singh	80000
76766	Crick	72000
83821	Brandt	92000
98345	Kim	80000

Project Operation

Example of the Projection: Query

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$$\Pi_{A,\ C}(r)$$

Result?

Project Operation

Example of the Projection: Query

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$\Pi_{A,\ C}(r)$

Result?

A	C
α	1
α	1
β	1
β	2

=

A	C
α	1
β	1
β	2

Union Operation

- Notation: $r \cup s$
- Defined as:

$$r \cup s = \{t \mid t \in r \text{ or } t \in s\}$$

- For $r \cup s$ to be valid:
  1. r, s must have the same **arity** (same number of attributes)
  2. The attribute domains must be **compatible** (2nd column of r deals with the same type of values as does the 2nd column of s)

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

A	B
α	1
α	2
β	1
β	3

r ∪ s

Union Operation

Example of the Union: Query

- Find all courses taught in Fall 2017 semester, or Spring 2018 semester, or both
- Query?

Union Operation

Example of the Union: Query

Result?

- Find all courses taught in Fall 2017 semester, or Spring 2018 semester, or both
- Query?

$$\Pi_{course\_id}(\sigma_{semester="Fall" \land year=2017}(section)) \cup$$

$$\Pi_{course\_id}(\sigma_{semester="Spring" \land year=2018}(section))$$

Union Operation

Example of the Union: Query

Result?

- Find all courses taught in Fall 2017 semester, or Spring 2018 semester, or both
- Query?

$$\Pi_{course\_id}(\sigma_{semester="Fall" \land year=2017}(section)) \cup$$

$$\Pi_{course\_id}(\sigma_{semester="Spring" \land year=2018}(section))$$

course_id
CS-101
CS-315
CS-319
CS-347
FIN-201
HIS-351
MU-199
PHY-101

Set Difference Operation

- Notation: $r - s$
- Defined as:

$$r - s = \{t \mid t \in r \textbf{ and } t \notin s\}$$

- Set differences must be taken between compatible relations:
  - r and s must have the same **arity**
  - attribute domains of r and s must be compatible

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

A	B
α	1
β	1

$r - s$

Set Difference Operation

Example of the Set Difference: Query

- Find all courses taught in Fall 2017 semester, but not in Spring 2018 semester
- Query?

Set Difference Operation

Example of the Set Difference: Query

- Find all courses taught in Fall 2017 semester, but not in Spring 2018 semester
- Query?

$$\Pi_{\text{course\_id}}(\sigma_{\text{semester}=\text{"Fall"} \wedge \text{year}=2017}(\text{section})) -$$

$$\Pi_{\text{course\_id}}(\sigma_{\text{semester}=\text{"Spring"} \wedge \text{year}=2018}(\text{section}))$$

Result?

Set Difference Operation

Example of the Set Difference: Query

- Find all courses taught in Fall 2017 semester, but not in Spring 2018 semester
- Query?

$$\Pi_{course\_id}(\sigma_{semester="Fall" \wedge year=2017}(section)) -$$

$$\Pi_{course\_id}(\sigma_{semester="Spring" \wedge year=2018}(section))$$

Result?

course_id
CS-347
PHY-101

Set-Intersection Operation

- Notation: $r \cap s$
- Defined as:

$$r \cap s = \{t \mid t \in r \textbf{ and } t \in s\}$$

- Assume:
  - r, s have the same *arity*
  - attributes of r and s are compatible
- Note: $r \cap s = r - (r - s)$

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

A	B
α	2

$r \cap s$

Set Intersection Operation

Example of the Set Intersection: Query

- Find the set of all courses taught in both Fall 2017 and Spring 2018 semesters
- Query?

Set Intersection Operation

Example of the Set Intersection: Query

- Find the set of all courses taught in both Fall 2017 and Spring 2018 semesters
- Query?

$$\Pi_{\text{course\_id}}(\sigma_{\text{semester="Fall"} \wedge \text{year}=2017}(\text{section})) \cap$$

$$\Pi_{\text{course\_id}}(\sigma_{\text{semester="Spring"} \wedge \text{year}=2018}(\text{section}))$$

Result?

Set Intersection Operation

Example of the Set Intersection: Query

- Find the set of all courses taught in both Fall 2017 and Spring 2018 semesters
- Query?

$$\Pi_{course\_id}(\sigma_{semester="Fall" \wedge year=2017}(section)) \cap$$

$$\Pi_{course\_id}(\sigma_{semester="Spring" \wedge year=2018}(section))$$

Result?

course_id
CS-101

Cartesian-Product Operation

- Notation: $r \times s$
- Defined as:

$$r \times s = \{tq \mid t \in r \textbf{ and } q \in s\}$$

- Assume that attributes of $r(R)$ and $s(S)$ are disjoint (That is, $R \cap S = \emptyset$)
- If attributes of $r(R)$ and $s(S)$ are not disjoint, then renaming must be used

A	B
α	1
β	2

r

C	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

s

A	B	C	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

r x *s*

Rename Operation

- Allows us to name, and therefore to refer to, the results of relational-algebra expressions
- Allows us to refer to a relation by more than one name
- Example:

$$\rho_x(E)$$

returns the expression E under the name X

- If a relational-algebra expression E has arity n, then

$$\rho_{x(A_1, A_2, \ldots, A_n)}(E)$$

returns the result of expression E under the name X, with the attributes renamed to $A_1, A_2, \ldots, A_n$

Division Operation

- The division operation is applied to two relations
- $R(Z) \div S(X)$, where $X \subseteq Z$. Let $Y = Z - X$ (and hence $Z = X \cup Y$); that is, let Y be the set of attributes of R that are not attributes of S.
- The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples t_R appear in R with $t_R[Y] = t$, and with
  - $t_R[X] = t_s$ *for every tuple* t_s in S
- For a tuple t to appear in the result T of DIVISION, the values in t must appear in R in combination with every tuple in S
- Division is a derived operation and can be expressed in terms of other operations

Division Operation Example

- Relations r, s:

  e.g.

  A is customer name
  B is branch-name

  1 and 2 here show two specific branch-names

  (Find customers who have an account in all branches of the bank)

r

A	B
α	1
α	2
α	3
β	1
γ	1
δ	1
δ	3
δ	4
ϵ	6
ϵ	1
β	2

s

B
1
2

- $r \div s$?

Division Operation Example

- Relations r, s:

  e.g.

  A is customer name
  B is branch-name

  1 and 2 here show two specific branch-names

  (Find customers who have an account in all branches of the bank)

r

A	B
α	1
α	2
α	3
β	1
γ	1
δ	1
δ	3
δ	4
ϵ	6
ϵ	1
β	2

s

B
1
2

- $r \div s$?

A
α
β

Division Operation Example (Multi-Attribute)

- Relations r, s:

  e.g.

  Students who have taken both "a" and "b" courses, with instructor "1"

  (Find students who have taken all courses given by instructor 1)

r

A	B	C	D	E
α	a	α	a	1
α	a	γ	a	1
α	a	γ	b	1
β	a	γ	a	1
β	a	γ	b	3
γ	a	γ	a	1
γ	a	γ	b	1
γ	a	β	b	1

s

D	E
a	1
b	1

- $r \div s$?

Division Operation Example (Multi-Attribute)

- Relations r, s:

  e.g.

  Students who have taken both "a" and "b" courses, with instructor "1"

  (Find students who have taken all courses given by instructor 1)

r

A	B	C	D	E
α	a	α	a	1
α	a	γ	a	1
α	a	γ	b	1
β	a	γ	a	1
β	a	γ	b	3
γ	a	γ	a	1
γ	a	γ	b	1
γ	a	β	b	1

s

D	E
a	1
b	1

- $r \div s$?

A	B	C
α	a	γ
γ	a	γ

Formal Definition of Relational Algebra

- A basic expression in the relational algebra consists of either one of the following:
  - A relation in the database
  - A constant relation
- Let E_1 and E_2 be relational-algebra expressions; the following are all relational-algebra expressions:
  - $E_1 \cup E_2$
  - $E_1 - E_2$
  - $E_1 \times E_2$
  - $\sigma_p(E_1)$, P is a predicate on attributes in E_1
  - $\Pi_S(E_1)$, S is a list consisting of some attributes in E_1
  - $\rho_x(E_1)$, x is the new name for the result of E_1

Tuple Relational Calculus

- A nonprocedural query language, where each query is of the form:

$$\{t \mid P(t)\}$$

- It is the set of all tuples t such that predicate P is true for t
- t is a tuple variable, $t[A]$ denotes the value of tuple t on attribute A
- $t \in r$ denotes that tuple t is in relation r
- P is a formula similar to that of the predicate calculus

Predicate Calculus Formula

- Set of attributes and constants
- Set of comparison operators: (e.g., $\leq, \geq, =, \neq, <, >$)
- Set of connectives: and ($\wedge$), or ($\vee$), not ($\neg$)
- Implication ($\Rightarrow$): $x \Rightarrow y$, if x is true, then y is true

$$x \Rightarrow y \equiv \neg x \vee y$$

- Set of quantifiers:
  - $\exists t \in r(Q(t))$ – "there exists" a tuple t in relation r such that predicate $Q(t)$ is true
  - $\forall t \in r(Q(t))$ – Q is true "for all" tuples t in relation r

Safety of Expressions

- It is possible to write tuple calculus expressions that generate infinite relations
- For example: $\{t \mid \neg(t \in r)\}$ results in an infinite relation if the domain of any attribute of relation r is infinite
- To guard against this problem, we restrict the set of allowable expressions to safe expressions
- An expression $\{t \mid P(t)\}$ in the tuple relational calculus is **safe** if every component of t appears in one of the relations, tuples, or constants that appear in P
  - NOTE: This is more than just a syntax condition
    - E.g., $\{t \mid t[A] = 5 \lor \text{true}\}$ is not safe — it defines an infinite set with attribute values that do not appear in any relation or tuples or constants in P

Domain Relational Calculus

- A nonprocedural query language equivalent in power to the tuple relational calculus
- Each query is an expression of the form:

$$\{x_1, x_2, \ldots, x_n \mid P(x_1, x_2, \ldots, x_n)\}$$

- $x_1, x_2, \ldots, x_n$ represent domain variables
- P represents a formula similar to that of the predicate calculus

Equivalent Notations in RA, TRC, and DRC

Select Operation

$R = (A, B)$

Relational Algebra:	$\sigma_{B=17}(r)$
Tuple Calculus:	$\{t \mid t \in r \wedge t[B] = 17\}$
Domain Calculus:	$\{\langle a, b\rangle \mid \langle a, b\rangle \in r \wedge b = 17\}$

Equivalent Notations in RA, TRC, and DRC

Project Operation

$R = (A, B)$

Relational Algebra: $\Pi_A(r)$

Tuple Calculus: $\{t \mid \exists p \in r(t[A] = p[A])\}$

Domain Calculus: $\{\langle a \rangle \mid \exists b(\langle a, b \rangle \in r)\}$

Equivalent Notations in RA, TRC, and DRC

Combining Operations

$R = (A, B)$

Relational Algebra:	$\Pi_A(\sigma_{B=17}(r))$
Tuple Calculus:	$\{t \mid \exists p \in r(t[A] = p[A] \wedge p[B] = 17)\}$
Domain Calculus:	$\{\langle a \rangle \mid \exists b(\langle a, b \rangle \in r \wedge b = 17)\}$

Equivalent Notations in RA, TRC, and DRC

Natural Join

$R = (A, B, C, D),\ \ S = (B, D, E)$

Relational Algebra: $r \bowtie s$

Tuple Calculus: $\{t \mid \exists p \in r \exists q \in s(t[A] = p[A] \wedge t[B] = p[B] \wedge$
$t[C] = p[C] \wedge t[D] = p[D] \wedge t[E] = q[E] \wedge$
$p[B] = q[B] \wedge p[D] = q[D])\}$

Domain Calculus: $\{\langle a, b, c, d, e\rangle \mid \langle a, b, c, d\rangle \in r \wedge \langle b, d, e\rangle \in s\}$

Equivalent Notations in RA, TRC, and DRC

Union

$R = (A, B, C) \qquad S = (A, B, C)$

Relational Algebra: $r \cup s$

Tuple Calculus: $\{t \mid t \in r \vee t \in s\}$

Domain Calculus: $\{\langle a, b, c\rangle \mid \langle a, b, c\rangle \in r \vee \langle a, b, c\rangle \in s\}$

Equivalent Notations in RA, TRC, and DRC

Intersection

$R = (A, B, C) \qquad S = (A, B, C)$

Relational Algebra: $r \cap s$

Tuple Calculus: $\{t \mid t \in r \wedge t \in s\}$

Domain Calculus: $\{\langle a, b, c\rangle \mid \langle a, b, c\rangle \in r \wedge \langle a, b, c\rangle \in s\}$

Equivalent Notations in RA, TRC, and DRC (7)

Set Difference

$R = (A, B, C) \qquad S = (A, B, C)$

Relational Algebra: $r - s$

Tuple Calculus: $\{t \mid t \in r \land t \notin s\}$

Domain Calculus: $\{\langle a, b, c\rangle \mid \langle a, b, c\rangle \in r \land \langle a, b, c\rangle \notin s\}$

Equivalent Notations in RA, TRC, and DRC

Cartesian/Cross Product

$R = (A, B) \qquad S = (C, D)$

Relational Algebra: $r \times s$

Tuple Calculus: $\{t \mid \exists p \in r \exists q \in s(t[A] = p[A] \wedge t[B] = p[B] \wedge$
$t[C] = q[C] \wedge t[D] = q[D])\}$

Domain Calculus: $\{\langle a, b, c, d\rangle \mid \langle a, b\rangle \in r \wedge \langle c, d\rangle \in s\}$

Equivalent Notations in RA, TRC, and DRC

Division

$R = (A, B) \qquad S = (B)$

Relational Algebra: $r \div s$

Tuple Calculus: $\{t \mid \exists p \in r \forall q \in s (p[B] = q[B] \Rightarrow t[A] = p[A])\}$

Domain Calculus: $\{\langle a \rangle \mid \langle a \rangle \in \Pi_A(r) \land \forall \langle b \rangle (\langle b \rangle \in s \Rightarrow \langle a, b \rangle \in r)\}$

Equivalent Notations in RA, TRC, and DRC

Use of the Universal Quantifier

$salary = (employee, salary\_amount)$

To find the maximum salary-amount:

(Extended) Relational Algebra:

$$max_{salary\_amount}(salary)$$

Tuple Calculus:

$$\{t \mid \forall p \in salary(p[salary\_amount] \leq t[salary\_amount])\}$$

Domain Calculus:

$$\{\langle s \rangle \mid \exists e(\langle e, s \rangle \in salary \land \forall e_1, s_1(\langle e_1, s_1 \rangle \in salary \Rightarrow s_1 \leq s))\}$$

Summery

- Discussed relational algebra with examples
- Introduced tuple relational and domain relational calculus
- Illustrated equivalence of algebra and calculus.

Acknowledgments/Contributions

- Some of the images utilized in these slides are subject to copyright - Abraham Silberschatz, Henry Korth, and S. Sudarshan. Database System Concepts. 6th Edition, McGraw-Hill Education
- Contributor 2024-25, 2025-26 - Yogesh K. Meena, IIT Gandhinagar